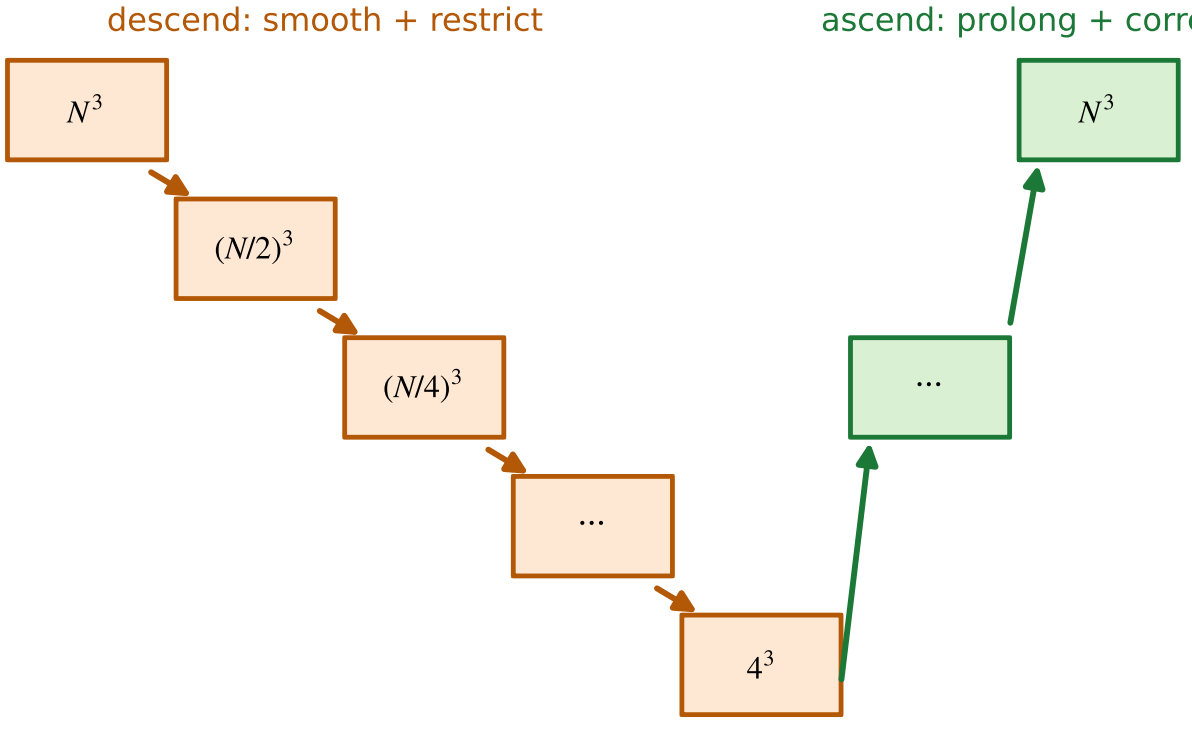
FMG multigrid Poisson solve



1. smooth (damped Jacobi): $\phi \leftarrow \phi + \omega D^{-1}(b - A\phi)$
2. residual: $r_h = b - A_h \phi_h$
3. restrict: $r_{2h} = \mathcal{R}\, r_h$ (full weighting)
4. coarse error eq.: $A_{2h}\, e_{2h} = r_{2h}$ (recurse)
5. prolong + correct: $\phi_h \leftarrow \phi_h + \mathcal{P}\, e_{2h}$
6. FMG: coarse solve warm-starts the next finer level;

one V-cycle/level $\Rightarrow$ discretisation accuracy

cost of a full FMG solve $\approx$ 2 V-cycles · matches the hydro's 2nd-order accuracy · every operation is a stencil $\Rightarrow$ GPU-native